

Aadarsha Gopala Reddy

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SUMMARY

M.S. Computer Science graduate (May 2026) from Washington University in St. Louis with hands-on experience across machine learning, data engineering, and full-stack development through research, teaching, and industry internships. Skilled in building end-to-end ML pipelines, real-time data systems, and production web applications with Python, PyTorch/TensorFlow, SQL, Spark/Kafka/Airflow, and modern JavaScript frameworks. Seeking software, data, or ML engineering roles.

SKILLS

Programming Languages: Java, C++, Python, R, HTML, CSS, JavaScript/TypeScript, PHP, C#, SQL, Rust.

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Node.js, Express.js, Vue.js, React, Socket.IO, Gemini/OpenAI API.

Tools & Technologies: Git, Tableau, Power BI, MySQL, AWS, MongoDB, Snowflake, Apache Airflow/Kafka/Spark, Streamlit, Microsoft 365, Google Workspace.

Core Competencies: Analytical Skills, Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Adaptability, Project Management.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.

EDUCATION

M.S. in Computer Science

August 2024 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- **GPA:** 3.67/4.00
- **Thesis:** Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data

B.A. in Computer Science and Data Analytics; Economics Minor

August 2020 - May 2023

Ohio Wesleyan University, Delaware, Ohio, USA

- **GPA:** 3.42/4.00; completed the four-year program in three years
- **Honors:** Golden Bishop Award, Mortar Board, Florence Leas Prize, Dean's List (Spring 2022, Spring 2023)

EXPERIENCE

Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale

January 2026 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Supported a graduate course on large-scale data pipelines using Python, Snowflake, SQL, Airflow, Spark, Kafka, and Flink, including fault tolerance and scaling.
- Conducted oral midterms structured as technical interviews and mentored three capstone teams building dashboards from open data sources.

AI Engineer Intern

June 2025 - August 2025

Crittero, Inc., Remote

- Built CritterCircle's (Crittero's pet platform) social media simulation, generating training data across seven personas via time-series modeling in TypeScript.
- Developed a TensorFlow/Keras recommendation model with a collaborative-filtering fallback and a configurable Python CLI for reproducible evaluation.

Graduate Assistant with the Taylor Family Center for Student Success

August 2024 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Organized engagement events for first-generation, limited-income (FLI) students (Welcome Dinner, First-Gen Week, End of Week Unwind), and managed the center's website and social media.
- Analyzed engagement, survey, and social media data in compliance with FERPA, developing process manuals to establish data evaluation best practices for staff.

Data Analyst Intern	June 2023 - May 2024
Lab714, Boca Raton, Florida, USA	
- Analyzed IoT environmental sensor data on AWS, building dashboards that helped clients cut pool maintenance costs by 20%. - Designed and engineered the UI for an upcoming all-in-one environmental sensor device in Figma and React.	

RESEARCH EXPERIENCE

Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data	August 2025 - May 2026
DRIVES Project, Washington University in St. Louis, St. Louis, Missouri, USA	
- Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving weekly driving features and integrating demographic covariates for CDR status prediction. - Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-group-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).	

LEADERSHIP

Graduate and Professional Student Council (GPSC) Washington University in St. Louis	
<i>Vice President of the Graduate-Professional Council (GPC) Chamber</i>	May 2025 - May 2026
- Managed the Council's unified Microsoft Teams workspace and website, establishing centralized communication hubs and maintaining up-to-date information on events, governance, and resources. - Represented the graduate student body to university leadership and supported the GPC President in advancing strategic council priorities. - Chaired the Constitution Committee (Aug 2025 - Feb 2026), coordinating stakeholder input to draft and ratify GPSC governing documents.	

PROJECTS

Datacenter Cooling Optimization using Deep Reinforcement Learning	August 2024 - December 2024
- Implemented multiple DRL algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations for datacenter cooling optimization, achieving up to 35.8% improvement over baseline controllers. - Developed a custom Sinergym environment simulating a small datacenter with stochastic weather conditions, focusing on the underserved small-to-mid-sized facility market.	
MLB Statcast Real-Time Data Pipeline	September 2025 - October 2025
- Built a lambda-architecture pipeline for MLB Statcast pitch-level data, using Apache Airflow for batch ingestion into Snowflake and a Kafka/Spark Streaming path for live, simulated data. - Developed a Streamlit dashboard with pitch-by-pitch game replay, interactive strike-zone visualizations, and pitcher/team analytics (velocity, spin rate, strikeout rate).	
Red-Blue Visual Auto Defender (VLM Security)	September 2025 - October 2025
- Built an automated red-blue teaming pipeline that generates visual prompt-injection (jailbreak) attack images and evaluates them against a vision-language model (Gemma-3-4b-it) using semantic response analysis. - Auto-generated deterministic, OCR-based Python defenses for each successful attack and validated them with accuracy/precision/recall metrics in an iterative refinement loop.	
Interactive Storybook	August 2024 - December 2024
- Built an interactive app with Vue.js, Node.js, and MongoDB, enabling AI-driven content generation and branching story narratives. - Integrated Google Gemini API and fal.ai for dynamic text/image generation, offering multiple story progression choices.	